Acta Biomaterialia 153 (2022) 364–373

Contents lists available at ScienceDirect

Acta Biomaterialia

journal homepage: www.elsevier.com/locate/actbio

Full length article

Extracellular matrix deformations of the porcine recurrent laryngeal nerve in response to hydrostatic pressure

Hirut G Kollech[b], Melissa R Chao[c], Amanda C Stark[d], Rebecca Z German[e], Randal C Paniello[f], Michael B Christensen[g], Julie M Barkmeier-Kraemer[h], Jonathan P Vande Geest[a,i,j,*]

[a] Department of Bioengineering, Swanson School of Engineering, University of Pittsburgh, Pittsburgh, PA, USA
[b] Computational Modeling and Simulation Program, University of Pittsburgh, Pittsburgh, PA, USA
[c] Department of Bioengineering, University of Utah, Salt Lake City, UT, USA
[d] National Center for Voice and Speech, University of Utah, Salt Lake City, UT, USA
[e] Department of Anatomy and Neurobiology, Northeast Ohio Medical University (NEOMED), Rootstown, OH, USA
[f] Department of Otolaryngology – Head and Neck Surgery, Washington University School of Medicine, St. Louis, MO, USA
[g] Department of Surgery, University of Utah, Salt Lake City, UT, USA
[h] Department of Surgery, Division of Otolaryngology–Head and Neck Surgery, University of Utah, UT, USA
[i] McGowan Institute for Regenerative Medicine, University of Pittsburgh, Pittsburgh, PA, USA
[j] Department of Mechanical Engineering and Material Science, Swanson School of Engineering, University of Pittsburgh, Pittsburgh, PA, USA

ARTICLE INFO

Article history:
Received 24 January 2021
Revised 22 August 2022
Accepted 15 September 2022
Available online 21 September 2022

ABSTRACT

Damage to the recurrent laryngeal nerve (RLN) caused by supraphysiological compression or tension imposed by adjacent tissue structures, such as the aorta, may contribute to onset of idiopathic unilateral vocal fold paralysis (iUVP) resulting in difficulty speaking, breathing, and swallowing. We previously demonstrated in adolescent pigs that the right RLN epineurium exhibits uniform composition of adipose tissue, with larger quantities along its length within the neck region in contrast to the left RLN that shows greater collagen composition in the thoracic region and greater quantities of adipose tissue in the neck region. In contrast, the epineurium in piglets was primarily composed of collagen tissue that remained uniform along the length of the left and right RLNs. Tensile testing of the left and right RLN in piglets and pigs showed associated differences in strain by RLN side and segment by age. The goal of this study was to investigate how external hydrostatic compression of the RLN affects the nerve's connective tissue and microstructure. RLN segments were harvested from the distal (cervical/neck) regions and proximal (subclavian for the right RLN, thoracic for the left RLN) regions from eight adolescent pigs and nine piglets. RLN segments were isolated and assessed under fluid compression to test hypotheses regarding epineurium composition and response to applied forces. Second harmonic generation (SHG) imaging of epineurial collagen was conducted at 0, 40, and 80 mmHg of compression. The cartesian strain tensor, principal strain (Eps1), and principal direction of the RLN collagen fibers were determined at each pressure step. Significantly larger values of the 1st principal strain occurred in the proximal segments of the pig left RLN when compared to the same segment in piglets ($p = 0.001$, pig = 0.0287 [IQR = 0.0161 - 0.0428], piglet = 0.0061 [IQR = 0.0033 - 0.0156]). Additionally, the median transverse strain E_{yy}) for the second pressure increment was larger in the right proximal segment of pigs compared to piglets ($p < 0.001$, pig = 0.0122 [IQR = 0.0033 - 0.0171], piglet = 0.0013 [IQR = 0.00001 - 0.0028]). E_{yy} values were significantly larger in the right proximal RLN versus the left proximal RLNs in pigs but not in piglets ($p < 0.001$). In contrast to piglets, histological analysis of pig RLN demonstrated increased axial alignment of epineurial and endoneurial collagen in response to compressive pressure. These findings support the hypothesis that the biomechanical response of the RLN to compressive pressure changed from being similar to being different between the right and left RLNs during development in the porcine model. Further investigation of these findings associated with age-related onset of idiopathic UVP may illuminate underlying etiologic mechanisms.

* Corresponding author.
E-mail address: jpv20@pitt.edu (J.P. Vande Geest).

https://doi.org/10.1016/j.actbio.2022.09.039
1742-7061/© 2022 Acta Materialia Inc. Published by Elsevier Ltd. All rights reserved.

Statement of significance

Damage to the recurrent laryngeal nerve (RLN) caused by compression imposed by the aorta may contribute to the onset of left-sided idiopathic unilateral vocal fold paralysis resulting in difficulty speaking, breathing, and swallowing. The goal of this study was to investigate how compression affects the connective tissue and microstructure of the RLN. We quantified the pressure induced deformation of the RLN using multiphoton imaging as a function of both location (proximal versus distal) and age (piglets, adolescent pigs). Our results demonstrate that the biomechanical response of the RLN to compression changes in the right versus left RLN throughout development, providing further evidence that the the left RLN is exposed to increasing dynamic loads with age.

© 2022 Acta Materialia Inc. Published by Elsevier Ltd. All rights reserved.

1. Introduction

Unilateral vocal fold paralysis (UVP) is a neurogenic disorder of the larynx associated with immobility of one vocal fold contributing to impaired airway safety during swallowing as well as abnormal voice production and breathing function [1–3]. UVP occurs due to varied etiologies including malignancy, central nervous system neuropathy, trauma, infection, and iatrogenic injury [4,5]. In the absence of a known etiology, UVP is classified as idiopathic (iUVP) [2,3,6]. Immobility of the impaired vocal fold in UVP is most commonly due to impaired function of the recurrent laryngeal nerve (RLN) that innervates the intrinsic muscles of the larynx [4,7-10].

Several studies demonstrate that left-sided UVP is approximately twice as prevalent as right-sided UVP and is associated with several forms of cardiothoracic disease including cardiac hypertrophy, thoracic aneurysm, and pulmonary artery engorgement [11–18]. The right and left-sided differences in prevalence in UVP may be explained by differing RLN length and location on each side of the body. Specifically, the right RLN bifurcates from the vagus nerve in proximity to the right subclavian artery before ascending to the posterior larynx. In contrast, the left RLN originates from the vagus nerve near the aortic arch within the thorax and travels around the inferior aortic arch before ascending toward the larynx within the tracheoesophageal groove, a much longer path [19] (Fig. 1A). Due to its anatomical course, the left RLN is in contact with several thoracic tissues and structures prior to reaching the larynx including the aorta, pulmonary artery, inferior and superior lobe of the left lung, esophagus, and trachea. As such, the left RLN especially offers greater opportunity for injury when neoplasms, trauma, or disease affect neighboring structures. In fact, the complicated anatomical course of this nerve also puts it at increased risk of damage in response to increased hydrostatic pressure from internal bleeding and edema following open heart surgery and thyroidectomy procedures [20] prevention and treatment.

One factor that has not been extensively evaluated as a potential etiology for the increased likelihood of impaired left RLN function is the biomechanical influence of adjacent large-diameter blood vessels. With aging, the aorta demonstrates gradual increase in diameter and reduced compliance [21,22]. We previously investigated the age- and gender-specific compliance of the aorta in patients with and without iUVP and showed that the aorta adjacent to the RLN was hypercompliant in those with iUVP compared to age- and gender-matched controls [23]. These findings indicated that higher compliance of the aorta in those with iUVP may impose a damaging level of supraphysiological stretch and/or compression on the left RLN resulting in impaired function. This hypothesis is also potentially supported by several studies in thoroughbred racehorses with iUVP associated with postmortem histopathological signs of chronic compression of the RLN in its proximal segments near the aorta, including loss of large-diameter axons and evidence of demyelination [24–26]. Chronic RLN compression resulting in nerve damage can potentially be influenced by the architecture of peripheral nerves and the several layers of connective tissue that protect and isolate axons against mechanical loading. The innermost layer, composed primarily of type I collagen, is called the endoneurium and provides support to axons [27]. Bundles of axons form nerve fascicles and are contained within the perineurium that is formed by concentric layers of modified fibroblasts [28–30]. The epineurium is the outermost layer of the peripheral nerve and encloses nerve fascicles and microvasculature [31]. Within the epineurium, type III collagen and elastin are the most commonly identified connective tissues with a predominantly longitudinal orientation. Adipose tissue is also present in the epineurium in varying quantities, depending on the anatomic location of the nerve [32]. Different components of peripheral nerve extracellular matrix function together and contribute to the mechanical properties of the tissue. For instance, researchers have suggested that epineurial collagen provides mechanical stiffness during stretch [33,34] whereas adipose tissue is attributed with protecting axons against compression [35].

Sunderland hypothesized that there is a connection between the biomechanical environment of the nerve (i.e., location) and its connective tissue composition. For example, he showed that the proportion of nerve epineurium is greater where a higher level of mechanical loading is applied [28,29,36]. In addition, Barkmeier et al. performed quantitative analysis of canine left RLN composition from its origin at the vagus to insertion into the posterior larynx [37]. This was the first study to demonstrate changes in the relative epineurial area from the proximal (i.e., near the aortic arch) to the distal (i.e., near the larynx) segment of the RLN [37]. This work was extended to demonstrate age-related changes in RLN epineurium in infant pigs (i.e. piglets) and juvenile pigs (3-4 month old pigs). Campbell et al. used stereology to study the proportion and composition of pig RLN connective tissue with aging. This study showed that piglets exhibited consistent proportions of collagen and adipose along the length of the left and right RLN [30]. In contrast, adolescent pigs exhibited significantly greater proportions of collagen tissue in the proximal than in the distal RLN as well as greater proportions of adipose connective tissue in the distal segments than in the proximal RLN [30]. Williams et al. mechanically evaluated different segments of piglet and adolescent pig left and right RLN using uniaxial tensile testing and found significant differences in the tangential modulus, strain energy density, and collagen fiber splay between various segments of the left RLN [38,39]. Williams et al. also showed that segments of the pig right RLN had different mechanical properties compared to the left RLN and that strain energy density was greater in piglet than adolescent pig RLN [38]. Consistent with Sunderland's theory, the above studies suggest that RLN connective tissue composition changes with aging in response to environmental loading conditions, and this is reflected in tensile testing. However, the RLN response to compressive forces has yet to be evaluated as a contributor to nerve changes. Sunderland previously proposed that nerves

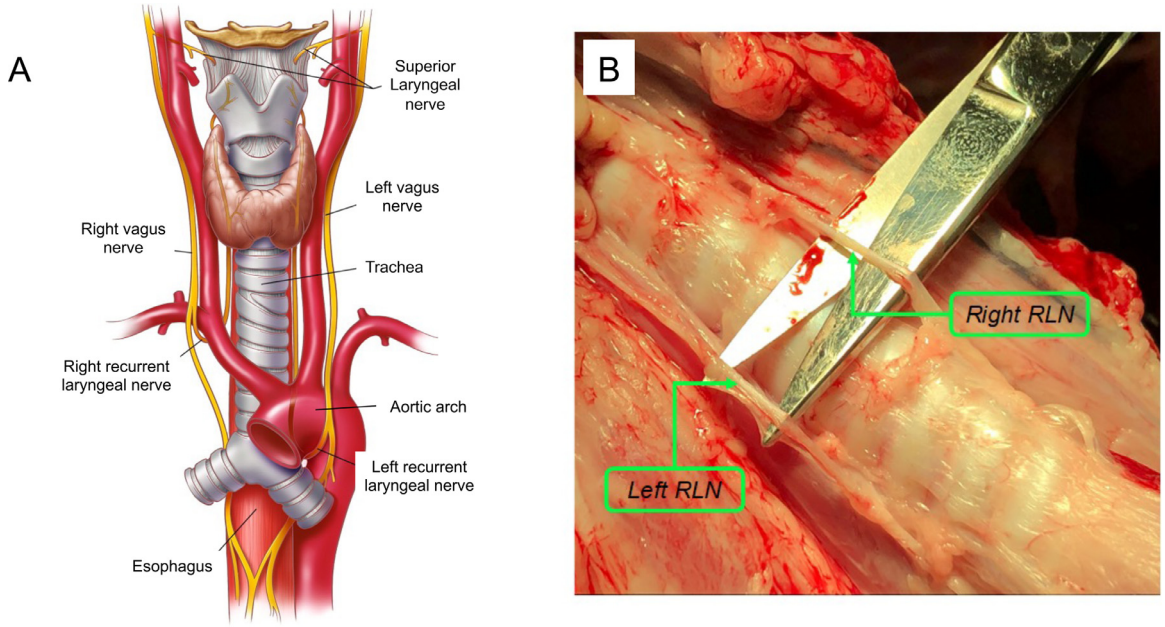


Fig. 1. A) Left and right RLN anatomy modified from [45], B) Left and right RLN of an adolescent pig. RLN Nerve diameter range from 1-2mm. The bottom right of this image is the cranial direction while the top left is caudal.

containing large quantities of epineurium primarily comprised of adipose tissue would redirect compressive forces around nerve fascicles to preserve their function [28,29,35-39]. However, prior work studying the rabbit tibial nerve demonstrated that epineurial arterial and infrafascicular vessel flow was impaired at 40-50 mmHg, and 80 mmHg of acute compression completely impaired intraneural blood flow within nerve epineurium resulting in permanently impaired nerve function [40]. Thus, compressive forces are anticipated to affect the microstructural organization of collagen and adipose tissues that protect nerve fibers within their fascicles, but also the blood vessels that maintain nerve function.

Given prior findings indicating RLN injury associated with supraphysiological levels of compression caused by a hypercompliant aorta in the region of the aortic arch, the purpose of this study was to investigate the biomechanical response of the RLN during hydrostatic compression. In order to conduct well-controlled mechanical testing of hydrostatic compression applied to fresh RLN tissues acquired from the thoracic cavity (proximal left RLN) and neck regions (proximal and distal right RLN and distal RLN), this study utilized a porcine animal model due to the inability to acquire fresh RLN tissues in humans across the lifespan. The porcine model of the RLN was chosen for this project given numerous prior studies demonstrating similarities between laryngeal anatomy, innervation and function as well as RLN anatomy between humans and porcine [41–44]. Second-harmonic generation (SHG) imaging was used to capture the deformation of superficial epineurial collagen in the RLN at different hydrostatic pressure levels in real time and determine the biomechanical response of the RLN as a function of location and age. A histological study was also performed on nerve samples following compression to examine the orientation of RLN collagen fibers as a function of compression and depth within the nerve.

2. Methods

2.1. Sample acquisition and preparation

Similar to prior anatomic and biomechanical studies comparing age-related changes within the porcine animal model, RLN was excised bilaterally from nine piglets (10 to 20-day old, five females) and eight pigs (3 to 4 months-old, four females) within 2 hrs. postmortem (Fig. 1B). All of these samples were acquired in an animal facility at the University of Pittsburgh. The animal study protocol was approved by the University of Pittsburgh's Institutional Animal Care and Use Committee (IACUC) based on the NIH guide for the care and use of laboratory animals.

RLN samples were stored in physiological saline at 4 °C. All nerve specimens were isolated from their insertion into the larynx (distal) to their origin at the main trunk of the vagus nerve (proximal). To maintain the orientation of the proximal and distal ends of the RLN, a segment of the vagus nerve trunk was harvested with the RLN tissue. Using microscissors and a dissecting microscope (Olympus-SZX7), extraneous external adipose and other adjacent connective tissues were gently removed from the RLN samples. The nerve was then divided into distal and proximal segments of approximately 2 cm in length. Due to the different anatomical locations of the proximal and distal portions of the left and right RLN, our comparisons focused on these segments for investigation. A small portion of the proximal and distal segments was harvested prior to mechanical testing and fixed in 10% formalin to serve as control samples. This enabled histological comparison of nerve diameter and collagen fiber orientation between control (untested) and mechanically tested samples. The experimental distal and proximal segments were mechanically tested after which all segments were stored in 10% formalin for 24 hrs prior to performing the histological analyses.

2.2. Compression test and imaging

The biomechanical response of the RLN under hydrostatic compression was studied using a custom-built compression chamber (Fig. 2A) designed to maintain pressure up to 80 mmHg. This maximal value was previously demonstrated to result in nerve ischemia resulting in permanent impaired function in rabbit peripheral nerves [40]. Both ends of each nerve specimen (~1mm) were glued (cyanoacrylate) to a coverslip to secure the sample during mechanical testing and imaging (Fig. 2C). The nerve was not stretched in the axial direction initially but was loose enough to

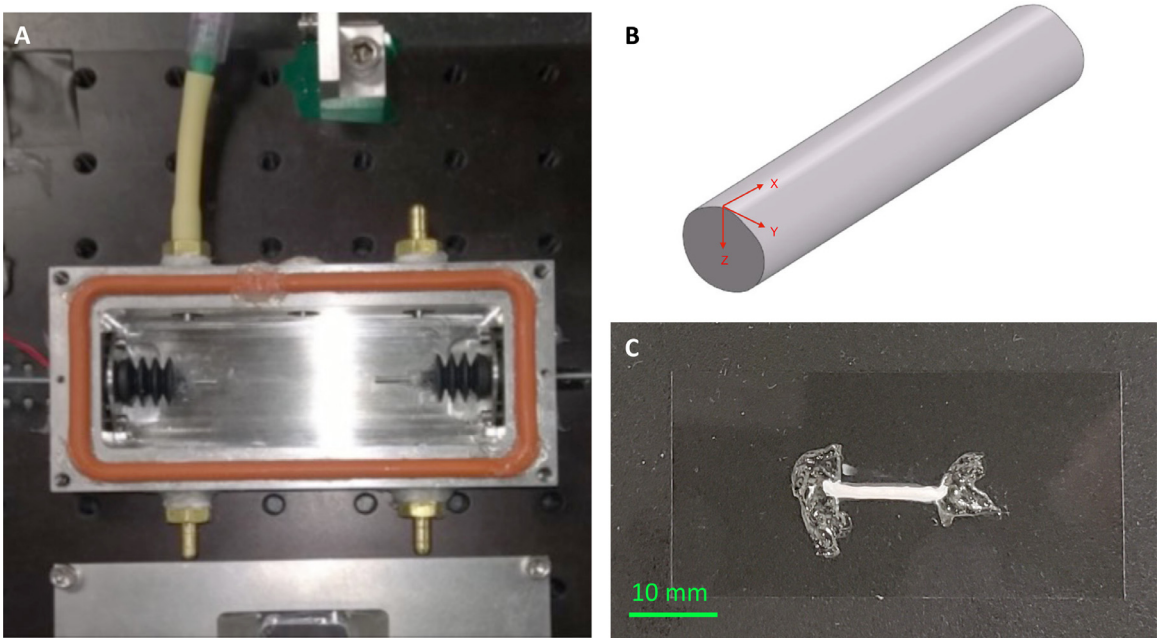


Fig. 2. A) A custom-built compression chamber was designed to maintain hydrostatic pressure up to 80 mmHg. B) The nerve specimen was glued to a coverslip. SHG Imaging was performed at the center point of the sample under compression. C) A Cartesian coordinate system was defined to investigate specimen strain field.

move freely and settle without any effect caused by adherence of the nerve ends to the coverslip using glue. The coverslip was then glued to the bottom of the compression chamber, and the lid was secured using flat head screws to maintain the pressure at various levels. A small hatch was designed on the chamber lid to image the specimen during mechanical testing. A hydrostatic column was used to create the desired pressure in the chamber (measured using a pressure gauge).

Based on prior work demonstrating important pressure levels causing change to nerve function, SHG imaging was performed at 0, 40, and 80 mmHg [40,46-48]. Our preliminary investigation showed that after 30 minutes, no significant change in the displacement field occurs; thus, at each pressure level, a 30-minute wait time prior to imaging was used to minimize the effect of creep. The samples were imaged using an Olympus BX51 upright laser-scanning microscope (Fig. 3A) that was coupled to a coherent 120-fs tunable pulsed titanium-sapphire laser (Chameleon ultra ii, Coherent Inc., Santa Clara, CA). The laser was centered at $\lambda = 780$ nm to efficiently visualize Type III collagen via SHG. Digital images were acquired with a plan-apochromatic lens with a magnification of 10X, numerical aperture = 0.3, a working distance of 3.8 mm, and a pixel size of 0.96 µm in the x- and y-direction and 3 µm in the z-direction with automatic focusing. The dimensions of each 2D image were 1042 × 1042 pixels, stored at 16 bits per pixel. An example of a mean intensity projection of the RLN is shown in Fig. 3B. Due to loss of signal at increased imaging depth, we excluded data deeper than 250 µm from the surface of the RLN in our displacement and strain calculations. Images used for analysis were also acquired from the center of each nerve specimen to avoid image acquisition from the extreme nerve segment boundaries that were glued to the coverslip.

2.3. Displacement and strain calculation

A digital volume correlation (DVC) algorithm was used to determine the displacement field for each sample across different pressure levels. Various algorithms [49–51] can be used to compute the displacement field at each pressure level. However, we needed a method to account for all types of displacement (i.e., translation, rotation, shear, and scale). In this study, we employed a non-rigid image registration technique based on Thirion's demons to calculate the displacement field. This algorithm utilized the concept of diffusing models (i.e., deformable grid), which is a robust and computationally efficient image registration approach [52,53]. Briefly, the DVC algorithm uses the pixel intensity and images texture to morph the undeformed image stack (e.g. at 0 mmHg) into the deformed one (e.g. 40 mmHg). One can assume that the procedure is similar to solving the diffusion equation numerically and iteratively where the diffusion coefficient is governed by image property. As the property of the image is not the same across the image stack, we can define a search domain to better navigate the DVC algorithm. We have verified and validated our DVC code by using predefined displacement fields. In addition, we ran experiments to carefully control the displacement and validate our code against expected experimental results [76].

A MATLAB (MathWorks, Natick, MA) code was developed based on a template matching algorithm [54] to compute the normalized cross-correlation coefficient values over many search windows between the deformed image stack acquired during the experiment (target stack) and the registered image stack (template stack) created by imposing the computed displacement field on the undeformed images. The template matching algorithm detected similarities in the texture and was independent of the voxel illumination. If two image stacks with the same size matched perfectly, the maximum cross-correlation coefficient (i.e. a value of one) occurred at the center of the target stack. We defined a normalized objective function $\omega(\zeta)$ to calculate rigidity of the deformable grid (ζ) that resulted in the optimized displacement field;

$$\omega(\zeta) = \upsilon[1 - \alpha(\zeta)] + [1 - \upsilon]\,\frac{\|\rho(\zeta) - C\|}{L} \tag{1}$$

where $\alpha(\zeta)$ is the value of the maximum of the cross-correlation coefficient, C is the location of the center of the target image stack, $\rho(\zeta)$ is the location of the maximum normalized cross-correlation coefficient, L is the largest dimension of the image stack, and υ is defined as a weight coefficient set to 0.4 for all image stacks. In an ideal case, when target and template stacks are identical, ω is equal to zero. A design of experiments study was performed to

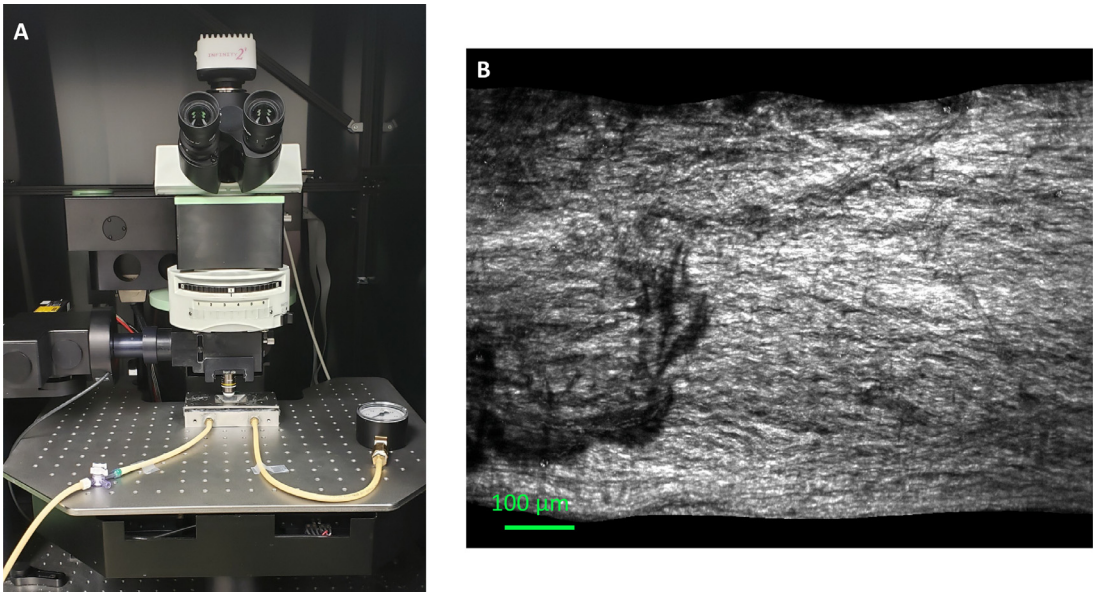


Fig. 3. A) Experimental setup under a multiphoton microscope to collect SHG signals. A pressure gauge was used to confirm the imposed pressure head. B) Mean intensity projection of a representative left RLN SHG image from a piglet at 0 mmHg.

find the value of ζ that minimized $\omega(\zeta)$ for each pressure increment within each sample. Upon completion of the displacement field calculation, a 3D Gaussian low-pass filter was applied to the displacement field to minimize the effect of noise and estimation error on the strain calculation [55].

The displacement results were then used to calculate strain; the relative displacement between particles in a material body. The Green strain tensor (**E**) is defined as:

$$\boldsymbol{E} = \frac{1}{2}\left(\boldsymbol{F}^T\boldsymbol{F} - \boldsymbol{I}\right), \quad \boldsymbol{F} = \frac{\partial \boldsymbol{u}}{\partial \boldsymbol{X}} + \boldsymbol{I} \tag{2}$$

$\boldsymbol{F}$ is the deformation gradient tensor, $\boldsymbol{I}$ is the identity matrix, and the displacement field is denoted as $\boldsymbol{u}$. All Cartesian components, 1st, 2nd, and 3rd principal green strain components were calculated. Principal strain values are eigenvalues of the strain tensor. A positive sign of the first principal strain shows the tensile nature of the strain in the first principal direction. A Cartesian coordinate system was defined as depicted in Fig. 2C. The X-axis was set in the axial direction of the nerve specimen, and Z-axis was defined from the nerve's superficial surface inward. The partial derivative of the displacement field and the strain tensor were calculated using a MATLAB code developed by Kroon [56,57]. Lastly, the minimum detectable difference for the displacement field components was determined to filter noise from the actual displacement field. In order to make sure the imaging noise is minimally affecting the results and filter this noise from the actual displacement field, this was done by imaging one sample twice without making any changes in the imaging setting. As pressure was maintained, the displacement gradient should be zero; however, due to imaging noise and round off/truncation error, a non-zero displacement field was observed. The average ± standard deviation of each displacement field component was used to define the noise level. Any strain value resulting from a displacement field with components smaller than the noise level was removed. We have done a considerable amount of work (as outlined in the methods section) to make sure that imaging noise minimally affects the results. It should be noted that the image shows the average of the value through the thickness. Noise levels of each displacement component were $u_x = 0.45 \pm 0.11$ µm, $u_y = 0.59 \pm 0.29$ µm and $u_z = 0.47 \pm 0.17$ µm.

2.4. Histological study

All nerves were imaged in cross-section following fixation. Cross-sectional areas were calculated using ImagePro Plus. For analysis of collagen fiber angles, tested and control samples were equilibrated in a 30% sucrose solution before sectioning using a Leica CM1950 cryostat (Leica, Bannockburn, IL). Two pigs and two piglets were used for qualitative analysis of fiber angles from surface to deep segments of sampled nerves. For this purpose, 100 µm thick longitudinal sections were serially collected and placed in wells. The depth of the section within the nerve was determined based on the distance from the exterior surface, regardless of nerve orientation (top vs. bottom). Type I collagen fibers were visualized via SHG imaging using a Bruker Ultima Multi-photon confocal microscope with a Nikon LWD 16x water immersion objective (numerical aperture of 0.8) and Prairie View Software. A Coherent Chameleon Vision IR laser was tuned to a wavelength of 800 nm, and detection was performed using non-descanned PMT detectors. SHG signals were discriminated from background autofluorescence using 460LP dichroic and 377/50 nm bandpass filters. XY pixel size was 0.81×0.81 µm, with a 7 µm z-step size. Collagen fiber angles were evaluated using a custom MATLAB code [58,59]. Fiber angles were normalized so that the mean fiber angle for each section was equal to zero (with the assumption that a zero angle is the direction along the length of the nerve). A total of 13,129,369 fiber angles were used for analysis.

2.5. Statistical analysis

A linear mixed model was used to analyze the compression test data. Pressure level (0-40 and 40-80 mmHg), nerve side (left and right), nerve section (distal and proximal), and age (piglet and pig) were considered as fixed effects while animal subjects were included as a random effect. Since the residuals of the strain data were not normally distributed, we used Box-Cox transformation to transform all data to normal [60]. After satisfying all requirements, we fit the transformed data to a linear mixed model and utilized 4-way ANOVA to identify significant differences with an alpha level of 5% using Bonferroni's correction to adjust for multiple comparisons [61]. Nerve cross-sectional areas were compared

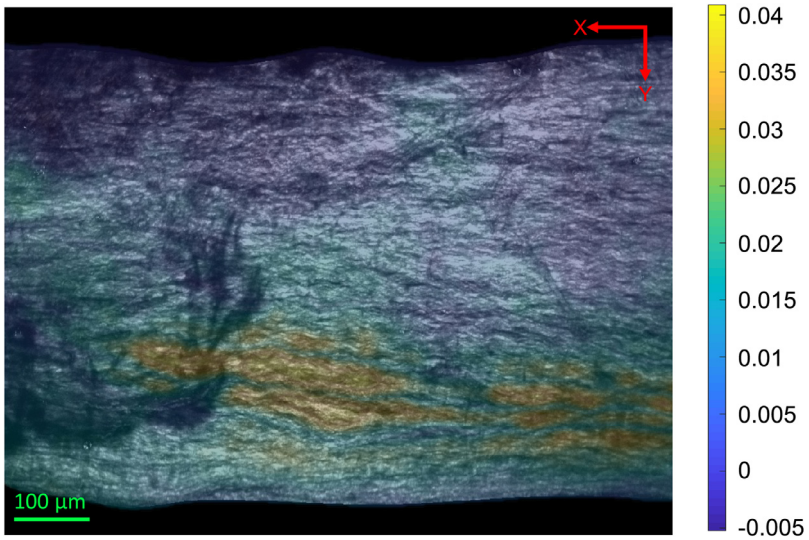


Fig. 4. Strain component in the axial direction (E_{xx}) averaged through the thickness (z-axis) for the distal section of a right piglet RLN at the first pressure increment (averaging was not performed in our statistical analysis).

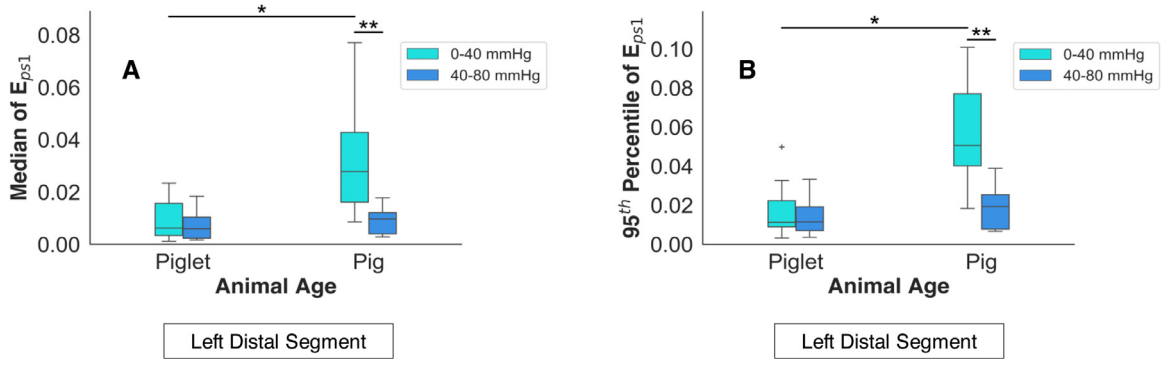


Fig. 5. Strain metrics for the left distal segment of the RLN as a function of animal age and pressure level. A) Median E_{ps1} values were significantly higher in pigs compared to piglets at the first pressure increment. These values were also higher in the pig group within the first pressure increment compared to the second. B) 95th percentile of the 1st principal strain showed a similar trend as the median of the E_{ps1}.

using a paired t-test. Collagen fiber angles were not statistically compared since only two animals per cohort were used.

3. Results

3D displacement field, strain components, and strain tensors were determined for all samples (Fig. 4). A significant main effect was identified for animal age group at the first pressure increment (i.e., 0 to 40 mmHg) with the median and 95th percentile (%95) of the 1st principal strain (E_{ps1}). Fig. 5A shows that the median of E_{ps1} for the left distal segment is significantly higher in pigs compared to piglets (*p-value = 0.004*). Also, the median of E_{ps1} was significantly larger at the first pressure increment compared to the second increment in pigs (*p-value = 0.018*).

Further, as depicted in Fig. 5B, a similar outcome occurred for %95 of the E_{ps1}. For the distal segment of the left RLN at the first pressure increment, %95 of E_{ps1} was significantly larger in pigs compared to piglets (*p-value = 0.003*). The pressure level was also significant for the %95 of E_{ps1} in the left distal segment of the RLN. Fig. 5B shows that within the pig group, %95 of E_{ps1} is larger at the first pressure increment compared to the second increment (*p-value = 0.016*). No significant differences were statistically demonstrated in the principal direction across the study groups. However, the average of the X component of the first principal direction was notably smaller than the other two components.

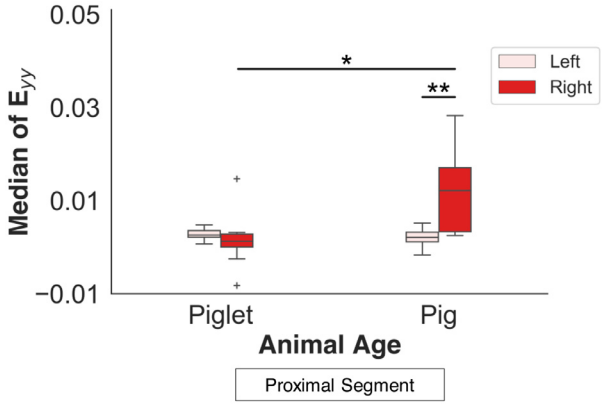


Fig. 6. Transverse strain (E_{yy}) in the second pressure increment for the proximal segment of the right and left RLN. Median E_{yy} values in the right RLN increased with age and were elevated compared to the left RLN in older animals.

A significant main effect (*p-value = 0.001*) was identified for the animal age group at the second pressure step with the median of the transverse strain (E_{yy}) being larger in the pig group than the piglets for the proximal section of the right RLN (Fig. 6). Moreover, the nerve side (left versus right RLN) showed a significant main effect for E_{yy}. That is, E_{yy} was significantly higher at the second

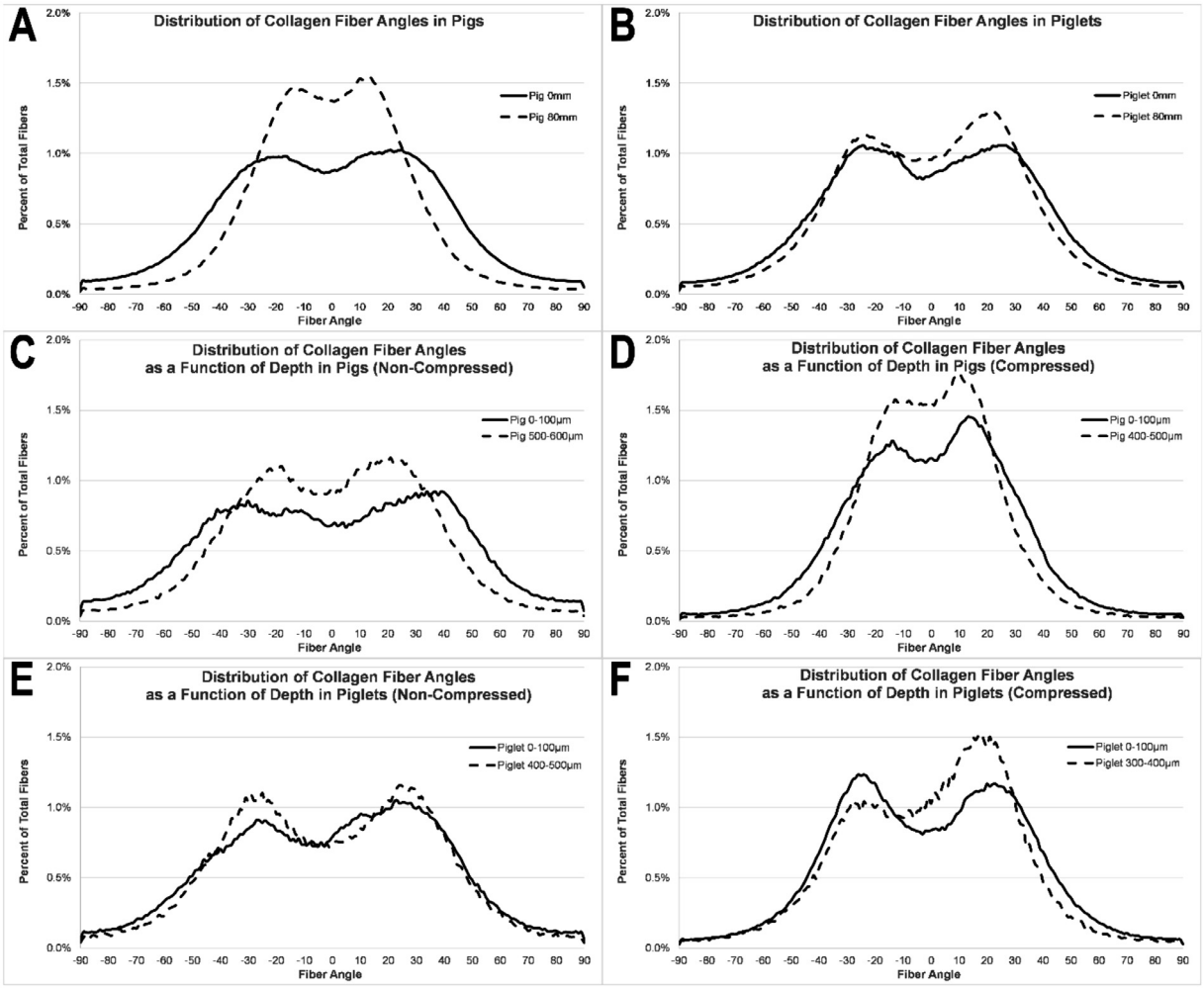


Fig. 7. Distribution of the collagen fiber angles. A) Distribution of fibers in pigs at 0 and 80 mmHg. B) Distribution of fibers in piglets at 0 and 80 mmHg. C) Distribution of fibers as a function of depth in non-compressed pig RLN. D) Distribution of fibers as a function of depth in compressed pig RLN. E) Distribution of fibers as a function of depth in non-compressed piglet RLN. F) Distribution of fibers as a function of depth in compressed piglet RLN.

Table 1
Diameter (mm) of RLNs from pig and piglet groups in both control and compressed nerve samples. Data are represented as mean ± SD, *p<0.05 compared to control samples.

	Control	Compressed
Pig	1.036 ±0.407	0.782 ±0.291*
Piglet	0.314 ±0.191	0.270 ±0.103

pressure level (40-80 mmHg) in the proximal section of the right pig RLN compared to the proximal segment of the left RLN (*p-value* < 0.001). The entire ANOVA statistical analysis results for all strain metrics are included in supplementary data.

Histological studies demonstrated a significant reduction in RLN diameter in the pig group following compression testing (*p-value* $= 0.026$, Table 1). However, a comparison between experimental and control samples in the piglet group did not show any statistically significant difference in RLN diameter (Table 1).

Qualitative evaluation of collagen fiber angles in a subgroup of the nerve samples denotes a similar finding. Collagen fiber angles in compressed pig nerves suggest an increased axial alignment distribution of fibers compared to non-compressed nerves (Fig. 7A). Results from piglet nerves suggest a minimal narrowing of the distribution (Fig. 7B). Qualitative evaluation of fiber angles as a function of nerve segment depth suggest that fibers deeper within the pig nerves tended to exhibit narrow distributions in both non-compressed and compressed nerves. In contrast, fiber angles within the piglet nerves suggest similar distributions regardless of depth (Fig. 7C-F).

4. Discussion

The findings of this study demonstrated several differences in porcine RLN biomechanical response across different age groups under compression. We showed that transverse strain (E_{yy}) is significantly higher in the proximal section of the right RLN compared to the same segment of left RLN in the pig group. These differences in RLN strain are attributed to variation in the microstructure and biomechanical properties of the RLN as a function of age and nerve location in pigs. Williams et al. assessed the microstructure and biomechanical response of porcine RLNs *using a uniaxial tensile test*. They demonstrated that the strain energy density was larger in the proximal segment of the left RLN compared to a similar segment on the right side [38]. Further, Campbell et al. conducted a study on RLN composition and showed that the proximal segment of the pig's right RLN at the level of the subclavian artery exhibited more adipose tissue and less collagen than found in the left proximal RLN that resides within the thorax near the aorta [30]. Thus, our finding of a higher transverse strain level in the proximal segment

of the right RLN could be attributed to the soft nature of adipose tissue found in higher quantities than in the same segment of the left RLN [62]. The latter would be consistent with Sunderland's hypothesis that there is an association between peripheral nerve biomechanical environment and connective tissue composition [28,29,36]. The proximal segment of the left RLN is adjacent to, and likely stretched and compressed by the aortic arch in addition to compression against the inferiorly located pulmonary artery during each cardiac cycle. Based on Sunderland's earlier findings, our finding of a lower transverse strain level in response to compression in the left proximal RLN aligns with anatomical findings of relatively greater collagen within the same epineurial cross sectional area [30]. The complex loading paradigm at the location of the left proximal RLN may necessitate a stiffer material (i.e., collagen versus adipose) that lowers strain values to preserve axons.

No significant differences were observed in the XZ, YZ or ZZ component of strain. However, principal values of strains are defined in the principal directions and are three dimensional entities, so out-of-plane strains are reported through the principal values. Principal values help us to better understand if tensile or compressive behaviors are intensified during the test and can better understand potential damage mechanisms. The entire statistical analysis results are reported in supplementary data.

We demonstrated that the median of first E_{ps1}) is significantly higher in pigs compared to piglets in the distal segment of the left RLN. Campbell et al. showed that the piglet epineurium primarily consisted of collagen whereas the pig epineurium was more heterogeneous being comprised of both [30]. Similarly, a study evaluating the human sciatic nerve anatomy showed that adipose connective tissue significantly increased [63] Thus, higher first principal strain values in the pig group could be attributed to the postnatal development of increased adipose connective tissue within the pig RLN compared to the piglet, as a biomechanical response to external forces experienced in the RLN's environment.

Campbell et al. demonstrated that piglet RLN tissue appeared more homogeneous within and between the left and right sides compared to pigs. They showed that the proportion of the epineurium and perineurium, as well as collagen and adipose content in piglets did not vary by location (i.e., left or right, proximal or distal) [30]. Homogeneity of the composition of the RLN within the piglet group implies that the biomechanical properties of the RLN in piglets would not vary at different locations. Thus, the biomechanical response of the nerve (i.e., strain field) within piglets should not be a function of nerve location. Consistent with prior anatomic findings and associated hypotheses, we did not find significant differences in the biomechanical response between the right versus left, or proximal versus distal RLN segments under compression within the piglet group.

Given that epineural collagen is primarily oriented longitudinally along both left and right RLNs [38], studies have shown that collagen contributes to mechanical stiffness during stretch and adipose tissue preserves the axons under compression [33–35]. Interestingly, our histological studies demonstrate a trend towards increasing orientation of collagen fibers under compression in the pig RLN. This increased organization of the collagen fibers may be necessitated by the decreased cross-sectional area experienced in these nerves as a result of compression; however, it is unclear if this increased organization plays a functional role in nerve protection. Notably, this trend towards increased collagen fiber orientation was not found in the piglet RLN. This may be due to the more homogenous nature and lack of development of the epineurial space in piglets, as was previously suggested [30].

Our results also suggest that collagen fibers deeper within the RLN in pigs may be more aligned with the nerve than more superficial fibers, under both non-compressed and compressed states (Fig. 7C-D). This may be due to the more limited relative epineurial space deeper within the nerve allowing for less randomization in angle. However, it remains unclear whether these apparent differences in fiber alignment translate to any functional differences offering a novel finding for future investigation. Depth did not appear to alter fiber angles within the piglet RLN, presumably because of the homogenous nature of the nerve in these immature animals [30]. Additional work is needed to determine if observed differences in fiber angle with depth in older animals are statistically significant.

Stretch-related injuries are cited by Sunderland as the most common peripheral nerve injury type [29]; however, compression injuries are also commonly attributed to nerve dysfunction [64,65]. While nerve compression may not involve rupture or tearing associated with stretch-induced injuries, loss of both motor and sensory functions is a possible outcome following supraphysiological compression [66,67]. The most common mechanisms for nerve damage due to compression include mechanical deformation and ischemia. Studies have shown that, even with short-term compression, ischemia can impart nerve damage by affecting large myelinated fibers [68–70]. In contrast, mechanical deformation is assumed to cause degenerative changes in the compressed nerve [71,72]. Various animal models of clinical (i.e., chronic) nerve compression have shown that nerve damage under chronic compression begins with the breakdown of the blood-nerve barrier, fibrosis, and subperineurial edema. In severe cases of injury, subsequent demyelination and axonal degeneration may occur, leading to permanent nerve damage [73–76]. Further, Rydevik et al. showed that supraphysiological external pressure could influence axonal transport and intraneural blood flow adversely [40]. It has also been demonstrated that external pressure between 20 to 30 mmHg can negatively affect venous flow within the epineurium, adversely influencing the nerve microenvironment [77].

Although extreme compression is not expected during repeated cyclic deformation of the left RLN around the aortic arch, changes in aorta diameter and compliance could potentially cause nerve damage over time. Watanabe et al. employed a rat forelimb model to study nerve injury as a result of continuous and repeated loading. They showed that constant stretch applied to the nerve using a small load (i.e., 2N) did not result in measurable histological, electrophysiological, or functional impairment. In contrast, a similar load applied in a cyclic manner (60 to 120 times per hour) was associated with significant histologic, electrophysiological and functional impairment of the nerve [78]. These findings suggest that the left RLN may be at risk for damage associated with repeated cyclic deformation during compression and stretch imposed by the aorta and pulmonary artery during each cardiac cycle. UVP onset is more prevalent in older populations [79] and was found to be associated with a level of compression or stretch slightly higher than normal due to hypercompliance of the aortic arch during each cardiac cycle. Thus, impaired left RLN function may be possible if the aortic arch remains hypercompliant over an extended period of time.

There are several important limitations in this study. Our imaging using a multiphoton microscope under pressure was depth limited. Each Z-stack of images covered about 250 μm, which is a small region and approximately 15% to 50% of the RLN diameter in pig and piglets. In addition, only SHG was imaged and as such the relative contribution of the other components besides collagen cannot be assessed.

We acknowledge that the field of view utilized in both SHG imaging with pressure and SHG imaging of histological sections was small relative to the overall size of the sample isolated. This limitation. especially as it relates to the heterogeneity of the response acquired, should be addressed in future studies. We also were constrained to impose hydrostatic pressure loading testing instead of compressing the sample against a soft tissue, as this

was a limitation of our current imaging and experimental setup. To overcome this limitation in the future, different methods of imaging, such as phase contrast μCT, can demonstrate the variation of the biomechanical response through the whole RLN diameter and eliminate the imaging depth limitation. Moreover, these new modalities may allow the quantification of the RLN deformation under more physiological *in vivo* conditions, for example in response to a simultaneous cyclic compression and tension. Finally, the time course of loading used in this manuscript should be carefully considered. In this work we applied a static hydrostatic load to the laryngeal nerves. In reality, this nerve is exposed to cyclic forces (both compression and stretch) and future studies should aim to better mimic this challenging biomechanical environment.

Our findings demonstrate that the biomechanical response of the RLN under mechanical compression differed significantly based on animal age, pressure level, RLN side, and location. These findings align with prior work using uniaxial testing of the RLN in pigs and piglets showing age-related differences aligned with developmental changes in epineurial composition. Considering that the onset of iUVP typically occurs later in life, it is plausible that gradual axonal degeneration in the RLN due to chronic supraphysiological levels of compression and tension originating from the pulsatile movement of a hypercompliant aortic arch may occur. Further investigation on the RLN microenvironment is required to better elucidate the association between large-diameter vessel mechanisms of damage to the RLN leading to iUVP. Specifically, future work should evaluate whether RLN function changes occur associated with abnormally larger cyclic diameter changes.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgment

Funding for this research was provided by the National Institute on Deafness and Other Communication Disorders (Grant R01-DC-011311: MPI: Barkmeier-Kraemer and Vande Geest). We would like to acknowledge Reza Behkam for his contributions to this work. He was partially supported by the NIH Cardiovascular Bioengineering Training Program (5T32HL076124).

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.actbio.2022.09.039.

References

[1] L. Mu, S.J.T.L. Yang, An experimental study on the laryngeal electromyography and visual observations in varying types of surgical injuries to the unilateral recurrent laryngeal nerve in the neck, Laryngoscope 101 (7) (1991) 699–708.
[2] L.N. Kelchner, et al., Etiology, pathophysiology, treatment choices, and voice results for unilateral adductor vocal fold paralysis: a 3-year retrospective, J. Voice 13 (4) (1999) 592–601.
[3] D.W. Chen, et al., Routine computed tomography in the evaluation of vocal fold movement impairment without an apparent cause, Otolaryngol. Head Neck Surg. 152 (2) (2015) 308–313.
[4] L.H.S. Rosenthal, M.S. Benninger, R.H.J.T.L. Deeb, Vocal fold immobility: a longitudinal analysis of etiology over 20 years, Laryngoscope 117 (10) (2007) 1864–1870.
[5] L. Sulica, The natural history of idiopathic unilateral vocal fold paralysis: evidence and problems, Laryngoscope 118 (7) (2008) 1303–1307.
[6] P. Tewari, S.K. Aggarwal, Combined left-sided recurrent laryngeal and phrenic nerve palsy after coronary artery operation, Ann. Thorac. Surg. 61 (6) (1996) 1721–1722.
[7] G. Ardito, et al., Revisited anatomy of the recurrent laryngeal nerves, Am. J. Surg. 187 (2) (2004) 249–253.
[8] D. Myssiorek, Recurrent laryngeal nerve paralysis: anatomy and etiology, Otolaryngol. Clin. North Am. 37 (1) (2004) 25–44.
[9] M.S. Benninger, J.B. Gillen, J.S.J.T.L. Altaian, Changing etiology of vocal fold immobility, Laryngoscope 108 (9) (1998) 1346–1350.
[10] T.S. Higgins, et al., Recurrent laryngeal nerve monitoring versus identification alone on post-thyroidectomy true vocal fold palsy: a meta-analysis, Laryngoscope 121 (5) (2011) 1009–1017.
[11] J.F.G. Escribano, et al., Ortner's syndrome and endoluminal treatment of a thoracic aortic aneurysm: a case report, Vasc. Endovascular Surg. 40 (1) (2006) 75–78.
[12] J.R. Day, R.K. Walesby, A spontaneous ductal aneurysm presenting with left recurrent laryngeal nerve palsy, Ann. Thorac. Surg. 72 (2) (2001) 608–609.
[13] V. Pressler, J. McNamara, Aneurysm of the thoracic aorta. Review of 260 cases, J. Thorac. Cardiovasc. Surg. 89 (1) (1985) 50–54.
[14] H. Bando, et al., Vocal fold paralysis as a sign of chest diseases: a 15-year retrospective study, World J. Surg. 30 (3) (2006) 293–298.
[15] E. Coscaron Blanco, J.L. Blanco Garcia, J.L. Gomez Gonzalez, [An uncommon cause for unilateral vocal fold paralisis: aortic aneurysms. Case report and literature review], An Otorrinolaringol Ibero Am 33 (5) (2006) 481–487.
[16] C. Wunderlich, et al., Ortner's syndrome or cardiovocal hoarseness, Intern. Med. J. 37 (6) (2007) 418–419.
[17] D.A. Dolowitz, C. Lewis, Left vocal cord paralysis associated with cardiac disease, Am. J. Med. 4 (6) (1948) 856–862.
[18] E.A. Spataro, D.J. Grindler, R.C. Paniello, Etiology and time to presentation of unilateral vocal fold paralysis, Otolaryngol. Head Neck Surg. 151 (2) (2014) 286–293.
[19] A. Monfared, G. Gorti, D.J.T.L. Kim, Microsurgical anatomy of the laryngeal nerves as related to thyroid surgery. 2002. 112(2): p. 386-392.
[20] Y. Jiang, et al., Prevention and treatment of recurrent laryngeal nerve injury in thyroid surgery, Int. J. Clin. Exp. Med. 7 (1) (2014) 101–107.
[21] B. Rylski, et al., Gender-related changes in aortic geometry throughout life, Eur J Cardiothorac Surg 45 (5) (2014) 805–811.
[22] D. Haskett, et al., Microstructural and biomechanical alterations of the human aorta as a function of age and location, Biomech Model Mechanobiol 9 (6) (2010) 725–736.
[23] R. Behkam, et al., Aortic arch compliance and idiopathic unilateral vocal fold paralysis, J Appl Physiol (1985) 123 (2) (2017) 303–309.
[24] J.E. Brakenhoff, et al., The prevalence of laryngeal disease in a large population of competition draft horses, Vet. Surg. 35 (6) (2006) 579–583.
[25] I. Duncan, I. Griffiths, R. Madrid, A light and electron microscopic study of the neuropathy of equine idiopathic laryngeal hemiplegia, Neuropathol. Appl. Neurobiol. 4 (6) (1978) 483–501.
[26] I. Duncan, J. Hammang, Ultrastructural observations of organelle accumulation in the equine recurrent laryngeal nerve, J. Neurocytol 16 (2) (1987) 269–280.
[27] V. Salonen, M. Roytta, J. Peltonen, The effects of nerve transection on the endoneurial collagen fibril sheaths, Acta Neuropathol 74 (1) (1987) 13–21.
[28] S.J.B. Sunderland, The connective tissues of peripheral nerves, Brain 88 (4) (1965) 841–854.
[29] S.S.J.M. Sunderland, N.O.J.o.t.A.A.o.E. Medicine, The anatomy and physiology of nerve injury. 1990. 13(9): p. 771-784.
[30] E.O. Campbell, et al., Developmental changes in the connective tissues of the porcine recurrent laryngeal nerve, J Anat 222 (6) (2013) 625–633.
[31] P.J.J.o.a. Thomas, The connective tissue of peripheral nerve: an electron microscope study, J. Anat. 97 (Pt 1) (1963) 35.
[32] A. Peters, S.L. Palay, H.d. Webster, The fine structure of the nervous system: the cells and their processes, Hoeber Medical Division, Harper & Row, 1970.
[33] C. Liu, C. Benda, F. Lewey, Tensile strength of human nerves: an experimental physical and histologic study, Arch. Neurol. Psychiatry 59 (3) (1948) 322–336.
[34] J. Haftek, Stretch injury of peripheral nerve: acute effects of stretching on rabbit nerve, J. Bone Joint Surgery. British volume 52 (2) (1970) 354–365.
[35] S. SUNDERLAND, The adipose tissue of peripheral nerves, Brain 68 (2) (1945) 118–122.
[36] S. Sunderland, Anatomical Features of Nerve Trunks in Relation to Nerve Injury and Nerve Repair: Chapter IV, Neurosurgery 17 (1970) 38–62.
[37] J.M. Barkmeier, E.S. Luschei, Quantitative analysis of the anatomy of the epineurium of the canine recurrent laryngeal nerve, J. Anatomy 196 (1) (2000) 85–101.
[38] M.J. Williams, et al., Differences in the microstructure and biomechanical properties of the recurrent laryngeal nerve as a function of age and location, J. Biomech. Eng.-Trans. Asme 136 (8) (2014).
[39] M.J. Alexander, J.M. Barkmeier-Kraemer, J.P.V. Geest, Biomechanical properties of recurrent laryngeal nerve in the piglet, Ann. Biomed. Eng. 38 (8) (2010) 2553–2562.
[40] B. Rydevik, G. Lundborg, U. Bagge, Effects of graded compression on intraneural blood flow: an in vivo study on rabbit tibial nerve, J. Hand Surg. [Am] 6 (1) (1981) 3–12.
[41] G.K. Gorti, et al., A preclinical model for laryngeal transplantation: anatomy and mucosal immunology of the porcine larynx, Transplantation 68 (11) (1999) 1638–1642.
[42] J.J. Jiang, J.R. Raviv, D.G. Hanson, Comparison of the phonation-related structures among pig, dog, white-tailed deer, and human larynges, Ann. Otol., Rhinol., Laryngol. 110 (12) (2001) 1120–1125.
[43] M.J. Knight, S.E. McDonald, M.A. Birchall, Intrinsic muscles and distribution of the recurrent laryngeal nerve in the pig larynx, Eur Arch Otorhinolaryngol 262 (4) (2005) 281–285.

[44] P. Stavroulaki, M. Birchall, Comparative study of the laryngeal innervation in humans and animals employed in laryngeal transplantation research, J. Laryngol. Otol. 115 (4) (2001) 257–266.
[45] Deslauriers, Anatomy of the neck and cervicothoracic junction, Thorac Surg Clin 17 (4) (2007) 529–547.
[46] D.J. Greenberg, G. Luciano, J.L. Frank, Transient recurrent laryngeal nerve palsy post-total thyroidectomy, Am Surg 85 (1) (2019) e36–e38.
[47] G. Lundborg, R. Myers, H. Powell, Nerve compression injury and increased endoneurial fluid pressure: a “miniature compartment syndrome”, J Neurol Neurosurg Psychiatry 46 (12) (1983) 1119–1124.
[48] P. Mattsson, J. Hydman, M. Svensson, Recovery of laryngeal function after intraoperative injury to the recurrent laryngeal nerve, Gland Surg 4 (1) (2015) 27–35.
[49] J.A. Maintz, M.A. Viergever, A survey of medical image registration, Med. Image Anal. 2 (1) (1998) 1–36.
[50] J.V. Hajnal, D.L. Hill, Medical Image Registration, CRC press, 2001.
[51] H. Lester, S.R.J.P.r. Arridge, A survey of hierarchical non-linear medical image registration. 1999. 32(1): p. 129-149.
[52] J.-P.J.M.i.a. Thirion, Image matching as a diffusion process: an analogy with Maxwell’s demons. 1998. 2(3): p. 243-260.
[53] X. Pennec, P. Cachier, N. Ayache, Understanding the “demon’s algorithm”: 3D non-rigid registration by gradient descent, International Conference on Medical Image Computing and Computer-Assisted Intervention, Springer, 1999.
[54] J.P. Lewis, Fast normalized cross-correlation. in Vision interface. 1995.
[55] K. Dabov, et al., Image denoising with block-matching and 3D filtering, Image Processing: Algorithms and Systems, Neural Networks, and Machine Learning, International Society for Optics and Photonics, 2006.
[56] D.-J. Kroon, Image and Point based Registration. 2016.
[57] K.Z. Abd-Elmoniem, M. Stuber, J.L. Prince, Direct three-dimensional myocardial strain tensor quantification and tracking using zHARP, Med. Image Anal. 12 (6) (2008) 778–786.
[58] S. Ram, et al., Three-dimensional segmentation of the ex-vivo anterior lamina cribrosa from second-harmonic imaging microscopy, IEEE Trans. Biomed. Eng. 65 (7) (2017) 1617–1629.
[59] E.A. Sander, V.H. Barocas, Comparison of 2D fiber network orientation measurement methods, J. Biomed. Mater. Res. Part A 88 (2) (2009) 322–331.
[60] G.E. Box, D.R. Cox, An analysis of transformations, J. R. Statis. Soc. 26 (2) (1964) 211–243.
[61] RStudio, RStudio: integrated development environment for R. 2018.
[62] N. Alkhouli, et al., The mechanical properties of human adipose tissues and their relationships to the structure and composition of the extracellular matrix, Am J Physiol Endocrinol Metab 305 (12) (2013) E1427–E1435.
[63] U.Z. Sladjana, J.D. Ivan, S.D. Bratislav, Microanatomical structure of the human sciatic nerve, Surg. Radiol. Anat. 30 (8) (2008) 619–626.
[64] L.M. Fortier, et al., An update on peroneal nerve entrapment and neuropathy, Orthop. Rev. (Pavia) 13 (2) (2021) 24937.
[65] M. Terrelonge Jr., L. Rosow, Common focal neuropathies in the hospitalized patient, Neurol. Clin. 40 (1) (2022) 175–190.
[66] M.G. Burnett, E.L. Zager, Pathophysiology of peripheral nerve injury: a brief review, Neurosurg Focus 16 (5) (2004) 1–7.
[67] R.M. Menorca, T.S. Fussell, J.C. Elfar, Nerve physiology: mechanisms of injury and recovery, Hand Clin. 29 (3) (2013) 317–330.
[68] M.S. Schickendantz, S. Yalcin, Conditions and injuries affecting the nerves around the elbow, Clin. Sports Med. 39 (3) (2020) 597–621.
[69] P.G. T Lewis, P Rothschild, Centripetal paralysis arising out of arrested bloodflow to the limb, Heart 16 (1931) 1–32.
[70] A.S. Burahee, et al., Cubital tunnel syndrome, EFORT Open Rev. 6 (9) (2021) 743–750.
[71] J. Ochoa, et al., Nature of the nerve lesion caused by a pneumatic tourniquet, Nature 233 (5317) (1971) 265.
[72] J. Ochoa, T. Fowler, R.W. Gilliatt, Anatomical changes in peripheral nerves compressed by a pneumatic tourniquet, J. Anat. 113 (Pt 3) (1972) 433.
[73] A.L. Dellon, S.E. Mackinnon, Chronic nerve compression model for the double crush hypothesis, Ann. Plast. Surg. 26 (3) (1991) 259–264.
[74] M.M. Jack, B.W. Smith, R.J. Spinner, Neurosurgery for the neurologist: peripheral nerve injury and compression (what can be fixed?), Neurol Clin 40 (2) (2022) 283–295.
[75] S.E. Mackinnon, et al., A primate model for chronic nerve compression, J. Reconstr. Microsurg. 1 (03) (1985) 185–194.
[76] J. O’Brien, et al., A model of chronic nerve compression in the rat, Ann. Plast. Surg. 19 (5) (1987) 430–435.
[77] L.B. Dahlin, W.G. McLean, Effects of graded experimental compression on slow and fast axonal transport in rabbit vagus nerve, J. Neurol. Sci. 72 (1) (1986) 19–30.
[78] M. Watanabe, et al., The implication of repeated versus continuous strain on nerve function in a rat forelimb model, J. Hand Surg. [Am] 26 (4) (2001) 663–669.
[79] S.M. Coyle, B.D. Weinrich, J.C. Stemple, Shifts in relative prevalence of laryngeal pathology in a treatment-seeking population, J. Voice 15 (3) (2001) 424–440.